CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech. EE Sem - VI Examination, December 2013

EE - 307 Electrical Power System -III

Date: 04/12/2013, Wednesday **Time: 1.30 p.m. to 4.30 p.m** **Maximum Marks: 70**

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1 Differentiate Between

i. Switch and Circuit Breaker 3

ii. Restriking voltage and Recovery voltage 2

iii. Making capacity and breaking capacity of circuit breaker 2

Q-2 A. With neat diagrams explain the current interrupting in AC circuit breaker. 7

B. From the following data of a 50 Hz generator: e.m.f. to neutral 7.5 KV(rms), reactance of generator and connected system 4 Ω, distributed capacitance to neutral 0.01μF, resistance is negligible. Find, 7

(a) The maximum voltage across the contacts of circuit breaker when it breaks a short circuit at instantaneous value of current is zero.

(b) The frequency of transient oscillation. and

(c) The average rate of rise of voltage up to the first peak of the oscillation.

OR

Q-2 A. Derive the equation of doubling effect for RL series circuit. 6

B. Explain the different way due to the ionization of gases occurs in the arc extinguishing chamber of circuit breaker. 6

C. Give reason: Magnetic blow out coil is provided in the air break circuit breaker. 2

Q-3 A. With neat sketch explain the construction and working of SF_6 Circuit Breaker. 7

B. Explain the term "Resistance switching". Also derive the equation of critical damping resistance. 7

OR

Q-3 A. With neat sketch explain the construction and working of Air Blast Circuit Breaker. 7

B. Explain the physical and chemical properties of SF_6 gas. 7

SECTION – II

Q -4 Explain the advantages of p.u. system. 7

Q-5 A A 3-phase generator rated 15 MVA, 13.2 KV has a solidly grounded neutral. It has X_1=30%, X_2=40% and X_0=5%. 7

i. what value of **reactance** must be placed in the generator neutral so that the fault current for L-G fault of zero fault impedance shall not be exceed the rated current.

ii. What value of **resistance** in neutral will serve the same purpose.

B Explain the power invariance theorem in connection with symmetrical components with necessary derivation 7

OR

Q-5 A Using appropriate interconnection of sequence networks, derive the equation for a line to ground fault in a power system with a fault impedance of Z_f. 7

B A part of power system is shown in figure 1. Find value of LL fault current when fault occurs at the motor terminal with fault resistance of 2 ohm. 7

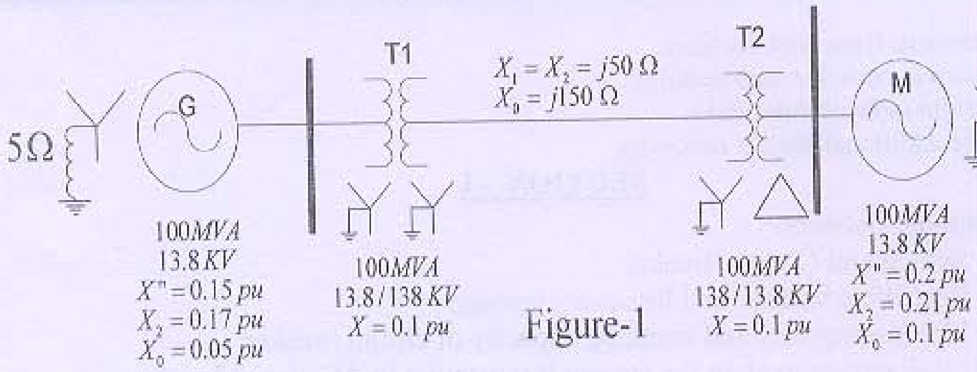


Figure-1

Q-6 A A 3-phase, 5 MVA, 6.6 KV alternator with a reactance of 8% is connected via a transformer to a feeder of series impedance (0.12+j0.48) ohm/phase/km. The transformer is rated at 3MVA, 6.6 KV/33 KV and has a reactance of 5%. Determine the fault current supplied by the generator operating under no load with a voltage of 6.9 KV, when a 3-phase symmetrical fault occurs at a point 15km along the feeder. 7

B Draw the p.u. reactance diagram for the part of the power system shown in the figure 2. The ratings of generators and transmission lines are shown in the figure and the rating of transformers are as under. 7

Transformer T_1 : 30 MVA, 6.6 Δ-115 Y KV, j0.1pu

Transformer T_2 : 15 MVA, 6.6 Δ-115 Y KV, j0.1pu

Transformer T_3 : Three single phase units each rated 10 MVA, 6.9/69 KV, j0.1pu

Draw an impedance diagram and all values in pu choosing a base of 30MVA, 6.6 KV in the circuit of generator-1.

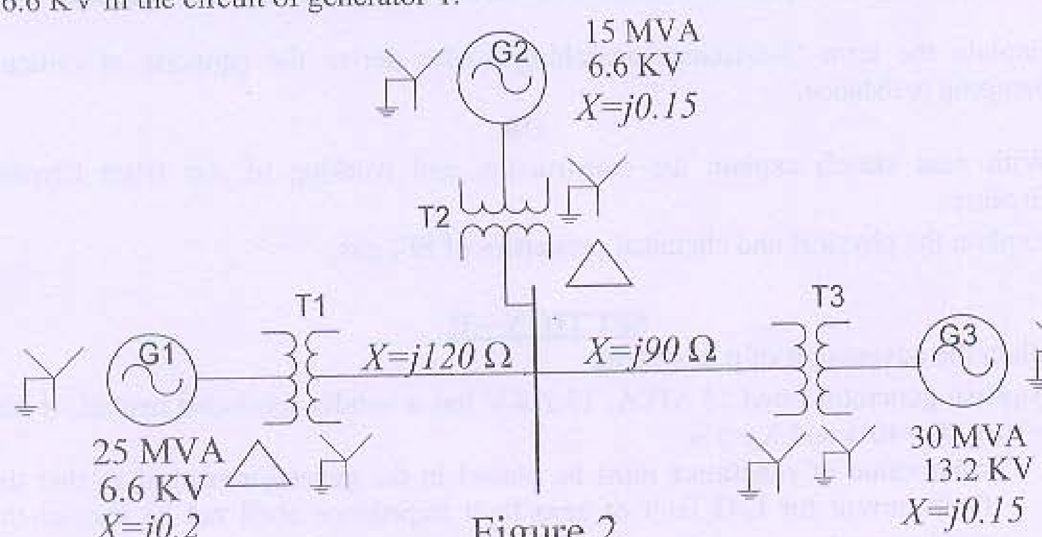
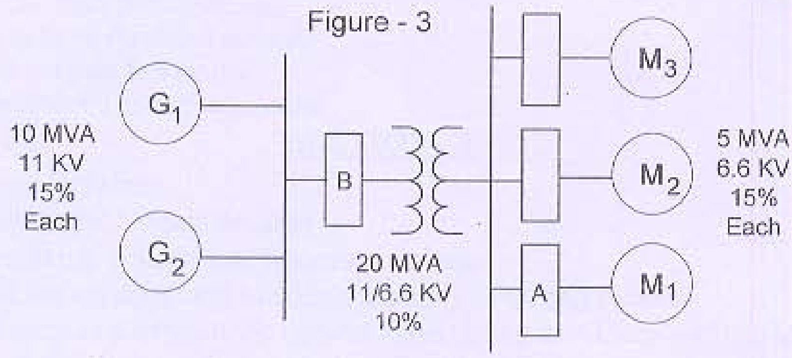


Figure 2

OR

Q-6 A Two 10 MVA, 11 KV generators with 15% sub-transient reactance is connected through a transformer to a bus that supplies three identical motors as shown in figure 3. Each motor has sub-transient reactance of 15% based on 5 MVA, 6.6 KV. Three phase rating of transformer is 20 MVA, 11/6.6 KV and leakage reactance is 10%. Bus voltage is 6.6 KV when a 3-Φ fault occurs at the 11 KV side terminal of transformer. Neglecting the prefault current, find out the current through breaker A and current through breaker B during fault. Take base MVA as per transformer rating. 7

Figure - 3



B One power system is operating at 33 KV is divided into sections A and B as shown in figure 4. Section A consist of three generators 15 MVA each having a reactance of 15% and section B is fed from the grid through a 75 MVA transformer of 8% reactance. The circuit breakers have each a rupturing capacity of 750 MVA. Determine the reactance of the reactor to prevent the breakers being overloaded if a symmetrical short-circuit occurs on an outgoing feeder connected to A. Take base MVA as 75 MVA. 7

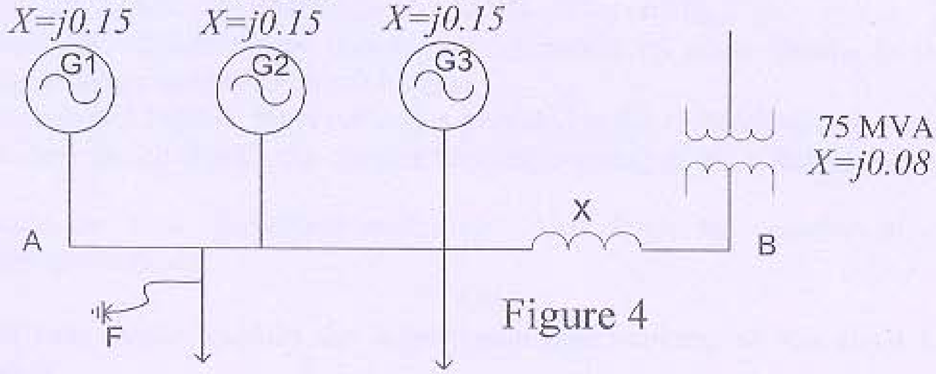


Figure 4

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